11. Wednesday, April 23. Divisibility (3.E)

Recall from last time:

Theorem 6: Let $a \in \mathbb{R}$. If $0 < a < 1$, then $0 < a^2 < a$ **Proof**: HW 4 □

Theorem 7: The integer 1 is the smallest positive integer.

Proof: Suppose for a contradiction there is some $x \in \mathbb{Z}$ with $0 < x < 1$.
Then the set $S = \{x \in \mathbb{Z} \mid 0 < x \text{ and } x < 1\}$ is non-empty.

Since $S \subseteq \mathbb{Z}^+$ is non-empty, by the well-ordering property, it has a minimum element, say y.
By closure of $\mathbb{Z}$, we have $y^2 \in \mathbb{Z}$. And by Thm 10.6 above, since $0 < y < 1$, we have $0 < y^2 < y < 1$. Therefore $y^2 \in S$ and $y^2 < y$, which contradicts that y is minimum. □

Technique (Infinite Descent) To prove that something in $\mathbb{Z}^+$ does not exist:
1) Assume for a contradiction the thing exists and take the minimum one
2) Find a smaller one. This is a contradiction.

Divisibility

Definition 1: We say that a non-zero integer m **divides** an integer n if there exists some integer q so that $n = m \cdot q$

In this case, we say that m is a **divisor** or a factor of n.

and n is a **multiple** of m.

The integer 0 is **not** a divisor of any integer.

Example: 7 divides 21, because $21 = 7 \cdot 3$. So 7 is a divisor of 21, and 21 is a multiple of 7.

Notation: If a divides b we write $a \mid b$ (\mid)

Theorem 2: For every nonzero integer a, we have $a \mid a$.

Proof: Let a be a nonzero integer. By the identity axiom, $a = a \cdot 1$, and since axiom 5 holds for $\mathbb{Z}$, we have $1 \in \mathbb{Z}$. Hence $a \mid a$ by definition of divisibility.

Theorem 3 Let $a,b,c \in \mathbb{Z}$ with $a \neq 0$ and $b \neq 0$. If $a \mid b$ and $b \mid c$, Then $a \mid c$.

Proof: Fix $a,b,c \in \mathbb{Z}$ with $a \neq 0$ and $b \neq 0$. Suppose $a \mid b$ and $b \mid c$.

By definition, this means There exist $q, p \in \mathbb{Z}$ such that $b = a \cdot q$ and $c = b \cdot p$.

Want $c = b \cdot k$ some k

By substitution, $c = b \cdot p = (a \cdot q) \cdot p$. By associativity $c = a \cdot (q \cdot p)$.

But $q \cdot p \in \mathbb{Z}$, because $\mathbb{Z}$ is closed.

Hence $a \mid c$ by definition of divisibility. □

Evens and odds Notation: $2 = 1+1$. So 2 is a positive integer different from 1.

Recall: An integer n is:
- Even if $n = 2k$ for some $k \in \mathbb{Z}$. i.e, $2 \mid n$
- Odd if $n = 2k+1$ for some $k \in \mathbb{Z}$.

Theorem 4 The integer 1 is odd

Proof: This is clear since $1 = 2 \cdot 0 + 1$.

Theorem 5 The integer 1 is not even.

Proof: Suppose for a contradiction That There exists $k \in \mathbb{Z}$ such That $1 = 2 \cdot k$.

We consider four cases That exhaustall possibilities, and show each leads To a contrad

Case 1: k is negative. If k is negative, Then $(-k)$ is positive, so $2 \cdot (-k)$ is positive by positive Closure. Since $2(-k) = -2k$, This means That $2k$ is negative. But $2k = 1$ is positive

by Thm 10.7. This contradicts Trichotomy, so case 1 is NOT possible.

Case 2: $K = 0$. If $K = 0$. Then $1 = 2K = 2 \cdot 0 = 0$. So $1 = 0$, which contradicts the identity axiom

Case 3: $K = 1$. If $K = 1$, Then $1 = 2K = 2 \cdot 1 = 2 = 1 + 1$.
By cancellation law, we get $0 = 1$, which is a contradiction.

Case 4: K is positive but $K \neq 1$. Since 1 is the smallest positive integer, we have $K > 1$.
By a HW exercise, since 2 is positive we get $2 \cdot K > 2 \cdot 1$, ie: $2K > 2$, Hence $1 > 2$.
This contradicts 1 is the smallest positive integer.

Any possibility for K leads to a contradiction, so it cannot exist, as desired $\square$

Technique (Breaking into cases) To prove something is true, prove that it is true in several specific cases, as long as these cases exhaust all possibilities.

Modular Arithmetic

Definition: Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}$. We say that a is congruent to b modulo n, and write $a \equiv b \pmod{n}$ if $n \mid (a-b)$.

(\equiv, \pmod{n})

Note: $a \equiv b \pmod{n}$ means $a - b = n \cdot k$ for some $k \in \mathbb{Z}$.

Example: "clock arithmetic"

Suppose it's 5:00. After 12 hours, the clock marks 5:00 again.

So for the clock, $5 = 5 + 12$. This equation is nonsense.

In modular arithmetic, we have $5 \equiv 17 \pmod{12}$

Moreover, for the clock all of the following are the same:

$$\ldots, -7, \underbrace{5}_{12}, \underbrace{17}_{12}, \underbrace{29}_{12}, \ldots$$

And $\{\ldots, -7, 5, 17, 29, \ldots\} = \{x \in \mathbb{Z} \mid x \equiv 5 \pmod{12}\}$

What time is it 39 hours after 5:00? It'll be 8:00 because $39 = 36 + 3$

Question: How is arithmetic modulo 2 related to evenness and oddness?

Evens: $\ldots -4, -2, 0, 2, 4, 6, 8, 10, \ldots$

Odds: $\ldots, -3, -1, 1, 3, 5, 7, \ldots$

So "Even" means congruent to 0 modulo 2
"odd" means congruent to 1 modulo 2.

Theorem 7: Let $n \in \mathbb{N}$, The relation $_ \equiv _ \pmod n$ is:

1) Reflexive: For all $a \in \mathbb{Z}$, $a \equiv a \pmod n$
2) Symmetric: For all $a, b \in \mathbb{Z}$, if $a \equiv b \pmod n$ Then $b \equiv a \pmod n$
3) Transitive: For all $a, b, c \in \mathbb{Z}$. if $a \equiv b \pmod n$ and $b \equiv c \pmod n$, Then $a \equiv c \pmod n$

Proof: HW 4 □

Note: This Theorem says That $_ \equiv _ \pmod n$ behaves like equality in some weaker sense.

Next: why modular "Arithmetic"

Theorem 8: Let $a, b, c, d \in \mathbb{Z}$ and $n \in \mathbb{N}$.

If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, Then:

1) $a + c \equiv b + d \pmod{n}$

2) $ac \equiv bd \pmod{n}$

3) $a^m \equiv b^m \pmod{n}$ for any $m \in \mathbb{N}$.

Example: 1) $29 \cdot 17 \equiv ? \pmod{11}$

Since $29 \equiv 7 \pmod{11}$ and $17 \equiv 6 \pmod{11}$ we have

$$29 \cdot 17 \equiv 7 \cdot 6 \equiv 42 \equiv 9 \pmod{11}$$

2) $(26)^5 \equiv ? \pmod{24}$

Since $26 \equiv 2 \pmod{24}$ $\quad 26^5 \equiv 2^5 \equiv 32 \equiv 8 \pmod{24}$

Proof of Thm 8: (1) and (3) HW!

(2) Fix $a, b, c, d \in \mathbb{Z}$, and $n \in \mathbb{Z}$. Suppose $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$.

By definition, $n \mid (a-b)$ and $n \mid (c-d)$. Hence, there exist $k_1, k_2 \in \mathbb{Z}$ such that

$$a - b = n k_1 \quad \text{and} \quad c - d = n k_2.$$

> Want: $ac \equiv bd \pmod{n}$
> ie: $n \mid (ac - bd)$

Multiplying $a - b = nk_1$ by c gives

$$ac - bc = n k_1 c$$

And $c - d = n k_2$ by b gives

$$bc - bd = n k_2 b$$

Adding these together, we get

$$ac - bd = n k_1 c + n k_2 b = n(k_1 c + k_2 b)$$

By closure of $\mathbb{Z}$, $k_1 c + k_2 b \in \mathbb{Z}$, so $n \mid (ac - bd)$, which means $ac \equiv bd \pmod{n}$

□

Division Theorem: For any $a \in \mathbb{Z}$ and $n \in \mathbb{N}$, there exist unique integers q and r such that

$$a = nq + r \quad \text{with} \quad 0 \leq r < n.$$

q is called the quotient and r is called the reminder

Note:

For $n = 2$, this in particular implies that any $a \in \mathbb{Z}$ is either odd ($r = 1$) or even ($r = 0$) but not both (r is unique)

Example: $a = 93$, $n = 6$. Then $q = 15$ and $r = 3$

$$93 = 15 \cdot 6 + 3$$

For $a = -16$, $n = 6$, we have $-16 = 6 \cdot (-3) + 2$ so $q = -3$, $r = 2$

Corollary: For any $a \in \mathbb{Z}$ and $n \in \mathbb{N}$, there is a unique integer r with $0 \leq r < n$ such that

$$a \equiv r \pmod{n}$$